

Advanced Cisco Networking Academy - Routing with Multi-Area OSPF

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Purpose

To successfully transmit information between devices, packets require a defined path to navigate through the network. Routing protocols, such as Open Shortest Path First (OSPF), play an important role in determining the optimal routes for these packets to reach their intended destinations efficiently. This lab explains the process of configuring OSPF for routing within a network that utilizes multiple areas, ensuring effective communication across different segments of a network.

Background

In the 1950s, in direct response to the Soviet Union's launch of Sputnik, President Dwight D. Eisenhower established the Advanced Research Projects Agency (ARPA). One of the agency's primary objectives was to explore and evaluate the feasibility of developing a large-scale computer network, paving the way for significant advancements in technology and communication.

In the early 1960's, researchers began experimenting with connecting two computers together. During this time, theoretical work, including resource sharing, protocol layering, and most notably packet-switching, was studied. Packet-switching, proposed by Paul Baran and Donald Davies, breaks data into small packets that travel independently through a network and are reassembled at the destination. Each packet carries a destination address which tells the devices in the network where each packet needs to go.

In October 1965, Lawrence G. Roberts and Thomas Marill connected an MIT TX-2 computer to an SDC Q-32 computer over a leased line, an early networking experiment that showed remote program access was possible. This achievement was a milestone in networking, acting as a steppingstone for the beginnings of ARPANET. With the vision of computer networking showing hope, the U.S. began to envision the military applications. The US wanted a communications system with no central core. The Department of Defense had concerns about technology and defense gaps between the US and the Soviet Union. Additionally, the military needed a communications system which could not be taken down by the destruction of one site. It needed to be resilient. While this goal was present, ARPANET's purpose was always more academic than military.

In 1969, the first message is sent between UCLA and Stanford, marking the project as a success. To start out, ARPANET had four nodes. Over the early 1970s ARPANET expanded across the U.S., including satellite links to Hawaii, and completed its first international connections to NORSAR, Norway, and University College London in 1973. This project grew and evolved to become the modern-day internet: a network of networks.

To organize the methods of routing packets across networks, protocols were developed. These protocols used algorithms to determine the best path for

incoming packets to get to their destination. RIP is one of the oldest distance-vector routing protocols. RIPv1 was standardized in RFC 1058 (1988). RIPv2 was later specified in RFC 2453 with support for subnet masks/VLSM and other extensions.

However, the most common routing protocol since its development is OSPF (Open Shortest Path First). OSPF was developed in the late 1980s to 1990s as an intra-domain link-state interior gateway protocol. It works by having each router in a network build a complete map of the topology through the exchange of information about each router's directly connected links. To determine the best path for a packet, OSPF uses the shortest path first (SPF) algorithm which attempts to find the shortest routes to any given destination. This works for both IPv4 and IPv6 networks as OSPF has multiple versions. OSPFv2 supports IPv4 networks, and OSPFv3 supports both IPv4 and IPv6 due to the Address Family functionality.

To configure OSPF, routers must be given both a Router and Process ID. Process IDs allow a router to run multiple instances of OSPF, each with its own settings and parameters. These IDs are seen only by the router they are assigned on, meaning they can be duplicated across the network. Router IDs give a router a specific identity in a network. This ID is public, so it must be unique. If it is not manually assigned, it will be given the value of the highest active loopback address and use the largest IP address of an active interface as a backup if no loopback interfaces are configured.

But, on a large network with many routers, each constructing their own topology table which must get updated anytime a link comes up or goes down, there can often be a flood of Link State Update packets that temporarily slow down a network. To avoid this lag, OSPF was developed with the ability to segment a network into multiple areas. This allows large networks to be organized into smaller sections, each acting independently of one another. It also reduces the impact of a link-state change because it no longer requires every router to update their link-state database - just the ones in the same area.

However, OSPF can be implemented not only on routers but also on multi-layer switches (MLS). An MLS is a versatile device that integrates the functionalities of both switches and routers. These advanced switches can connect networks and layers while also supporting multiple protocols such as IPv4, IPv6, Ethernet, and VLANs. Utilizing an MLS rather than a conventional router and switch pair often results in higher speeds, a more streamlined network design, and significant cost savings.

Lab Commands

```
Switch(config)# sdm prefer dual-ipv4-and-ipv6 default
```

Selects a Switching Database Manager (SDM) template on a switch to allow simultaneous support for both IPv4 and IPv6 traffic. This command must be followed by reloading the switch to utilize the feature.

```
Switch(config)# ip routing
```

Enables routing to be done on a multi-layer switch. This command must be given to enable IP addressing on the switch interfaces.

```
Router(config)# ipv6 unicast-routing
```

Allow routers to route IPv6 traffic.

```
Router(config)# router ospf <PID>
```

```
Router(config-router)# router-id <RID>
```

Configures the Process ID <PID> and the Router ID <RID> for IPv4.

```
Router(config)# router ospfv3 <PID>
```

```
Router(config-router)# router-id <RID>
```

Configures the Process ID <PID> and the Router ID <RID> for IPv6.

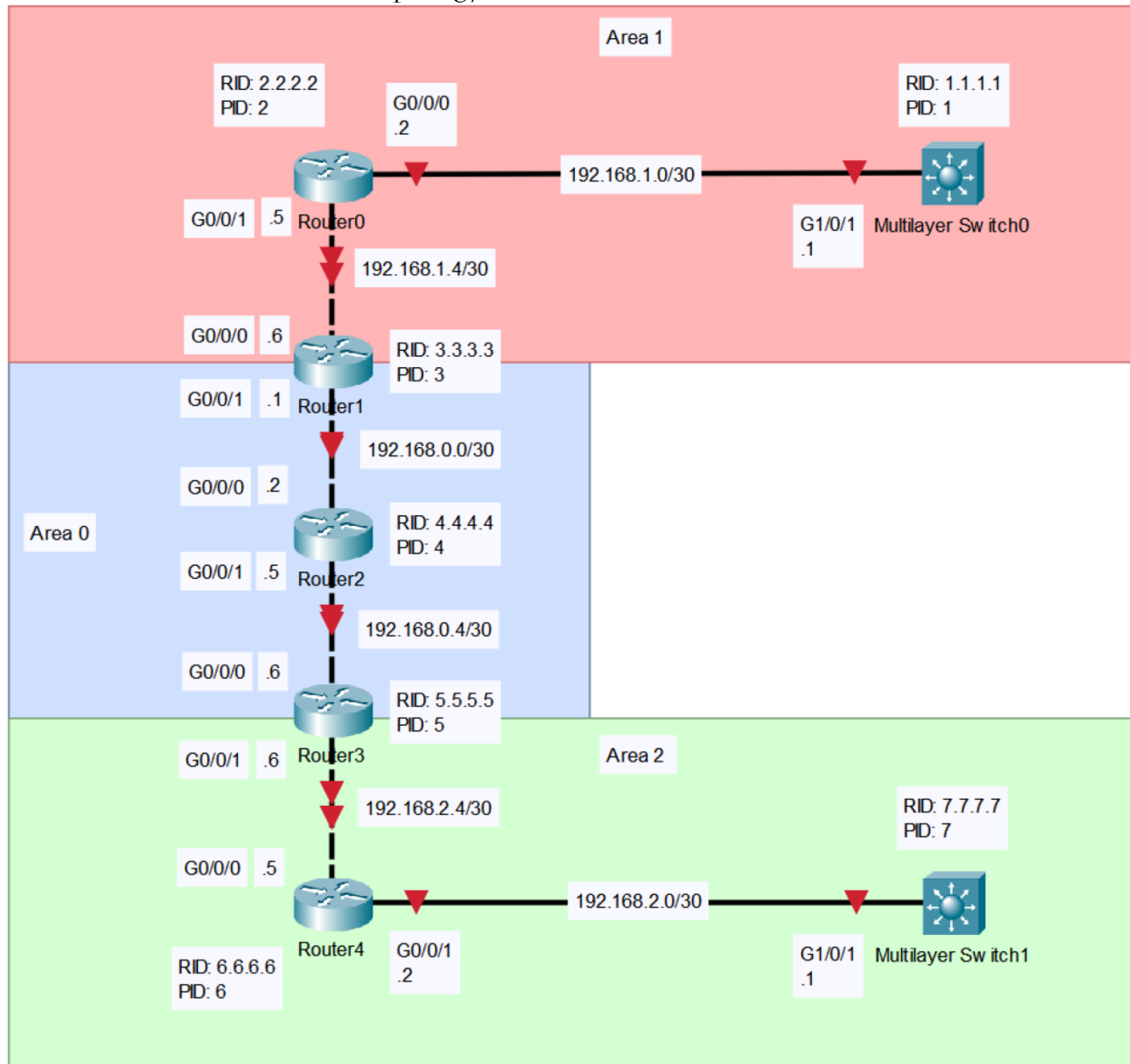
```
Router(config-if)# ip ospf <PID> area <area>
```

```
Router(config-if)# ipv6 ospf <PID> area <area>
```

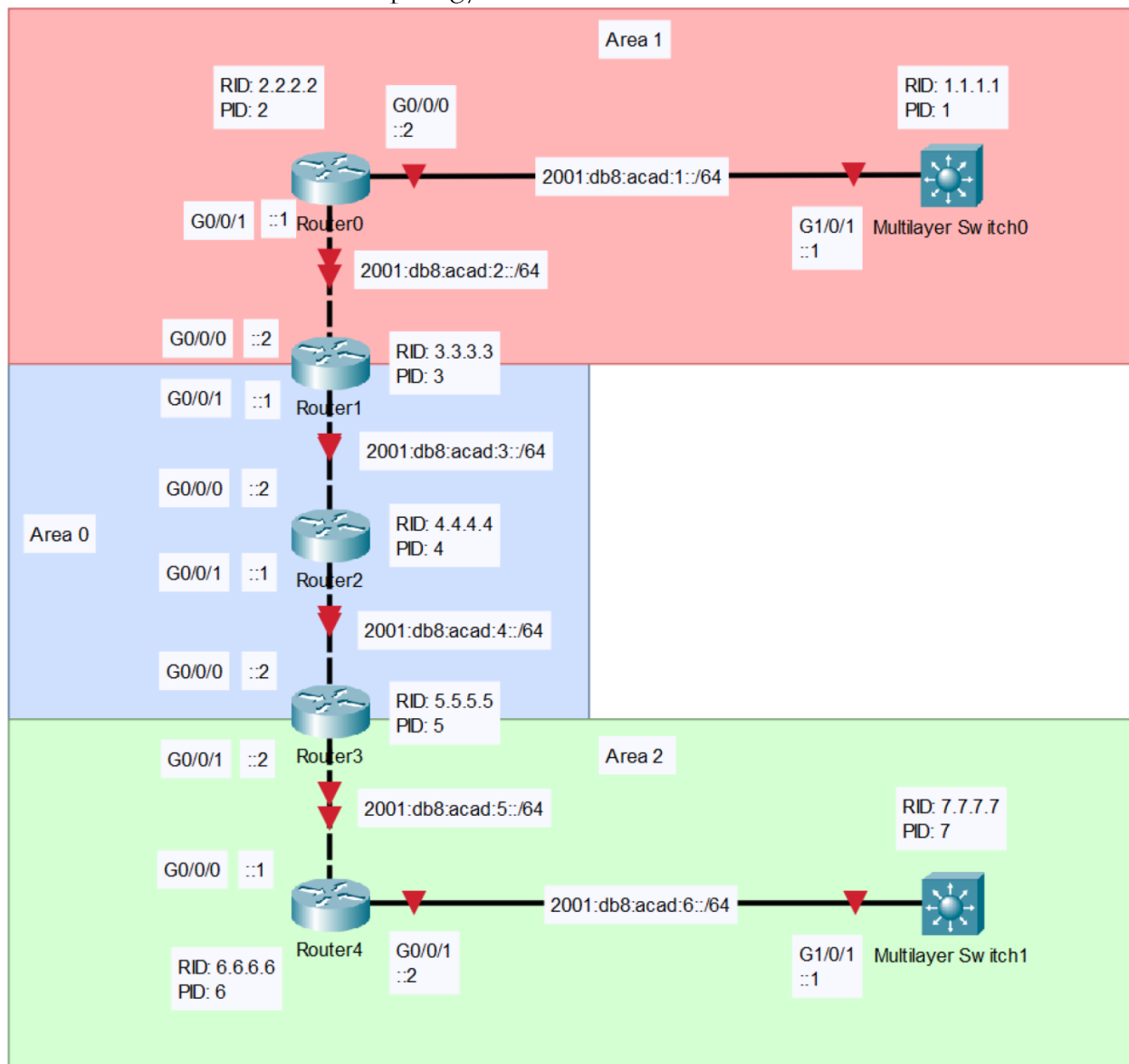
Enables an OSPF process on an interface for both IPv4 and IPv6 in a specified area.

Network Design

IPv4-Addressed Network Topology



IPv6-Addressed Network Topology



Configurations

Router0

```
version 16.9
hostname Router0
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 2
  router-id 2.2.2.2
router ospf 2
  router-id 2.2.2.2
interface GigabitEthernet0/0/0
  ip address 192.168.1.2 255.255.255.252
  ip ospf 2 area 1
  no shutdown
  ipv6 address 2001:db8:acad:1::2/64
  ipv6 ospf 2 area 1
interface GigabitEthernet0/0/1
  ip address 192.168.1.5 255.255.255.252
  ip ospf 2 area 1
  no shutdown
  ipv6 address 2001:db8:acad:2::1/64
  ipv6 ospf 2 area 1
```

Router1

```
version 16.9
hostname Router1
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 3
  router-id 3.3.3.3
router ospf 3
  router-id 3.3.3.3
interface GigabitEthernet0/0/0
  ip address 192.168.1.6 255.255.255.252
  ip ospf 3 area 1
  no shutdown
  ipv6 address 2001:db8:acad:2::2/64
  ipv6 ospf 3 area 1
interface GigabitEthernet0/0/1
  ip address 192.168.0.1 255.255.255.252
  ip ospf 3 area 0
  no shutdown
  ipv6 address 2001:db8:acad:3::1/64
  ipv6 ospf 3 area 0
```

Router2

```
version 16.9
hostname Router2
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 4
  router-id 4.4.4.4
router ospf 4
  router-id 4.4.4.4
interface GigabitEthernet0/0/0
  ip address 192.168.0.2 255.255.255.252
  ip ospf 4 area 0
  no shutdown
  ipv6 address 2001:db8:acad:3::2/64
  ipv6 ospf 4 area 0
interface GigabitEthernet0/0/1
  ip address 192.168.0.5 255.255.255.252
  ip ospf 4 area 0
  no shutdown
  ipv6 address 2001:db8:acad:4::1/64
  ipv6 ospf 4 area 0
```

Router3

```
version 16.9
hostname Router3
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 5
  router-id 5.5.5.5
router ospf 5
  router-id 5.5.5.5
interface GigabitEthernet0/0/0
  ip address 192.168.0.6 255.255.255.252
  ip ospf 5 area 0
  no shutdown
  ipv6 address 2001:db8:acad:4::2/64
  ipv6 ospf 5 area 0
interface GigabitEthernet0/0/1
  ip address 192.168.2.6 255.255.255.252
  ip ospf 5 area 2
  no shutdown
  ipv6 address 2001:db8:acad:5::2/64
  ipv6 ospf 5 area 2
```

Router4

```
version 16.9
hostname Router4
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
ipv6 unicast-routing
router ospfv3 6
  router-id 6.6.6.6
router ospf 6
  router-id 6.6.6.6
interface GigabitEthernet0/0/0
  ip address 192.168.2.5 255.255.255.252
  ip ospf 6 area 2
  no shutdown
  ipv6 address 2001:db8:acad:5::1/64
  ipv6 ospf 6 area 2
interface GigabitEthernet0/0/1
  ip address 192.168.2.2 255.255.255.252
  ip ospf 6 area 2
  no shutdown
  ipv6 address 2001:db8:acad:6::2/64
  ipv6 ospf 6 area 2
```

MLS0

```
version 16.9
hostname MLS0
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
This device has been configured for Mutli-Area OSPF $
ip routing
ipv6 unicast-routing
interface GigabitEthernet1/0/1
  no switchport
  ip address 192.168.1.1 255.255.255.252
  ip ospf 1 area 1
  no shutdown
  ipv6 address 2001:db8:acad:1::1/64
  ipv6 ospf 1 area 1
router ospfv3 1
  router-id 1.1.1.1
router ospf 1
  router-id 1.1.1.1
```

MLS1

```
version 16.9
hostname MLS1
banner motd $ UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED!
This device has been configured for Mutli-Area OSPF $
```

```

ip routing
ipv6 unicast-routing
interface GigabitEthernet1/0/1
  no switchport
  ip address 192.168.2.1 255.255.255.252
  ip ospf 7 area 2
  no shutdown
  ipv6 address 2001:db8:acad:6::1/64
  ipv6 ospf 7 area 2
router ospfv3 7
  router-id 7.7.7.7
router ospf 7
  router-id 7.7.7.7

```

Proof

To show that OSPF is running successfully in multiple areas on the network, there are a few commands that can be used as verification beyond a simple ping through the network.

MLS1#show ip route

Codes: L - local, C - connected, O - OSPF, IA - OSPF inter area

Gateway of last resort is not set

```

      192.168.0.0/30 is subnetted, 2 subnets
O   IA          192.168.0.0  [110/4]   via  192.168.2.2,    00:16:01,
GigabitEthernet1/0/1
O   IA          192.168.0.4  [110/3]   via  192.168.2.2,    00:16:01,
GigabitEthernet1/0/1
      192.168.1.0/30 is subnetted, 2 subnets
O   IA          192.168.1.0  [110/6]   via  192.168.2.2,    00:16:01,
GigabitEthernet1/0/1
O   IA          192.168.1.4  [110/5]   via  192.168.2.2,    00:16:01,
GigabitEthernet1/0/1
      192.168.2.0/24 is variably subnetted, 3 subnets, 2 masks
C           192.168.2.0/30    is      directly   connected,
GigabitEthernet1/0/1
L           192.168.2.1/32    is      directly   connected,
GigabitEthernet1/0/1
O           192.168.2.4/30

```

This command, and its IPv6 counterpart “show ipv6 route” reveal the routing table on the router or the MLS. The O next to the list of routes reveal that the MLS has successfully established routes using OSPF. Additionally, the IA next to the O code demonstrates the use of multiple areas.

MLS1#show ip ospf border-routers

```
OSPF Router with ID (7.7.7.7) (Process ID 7)
```

```
Base Topology (MTID 0)
```

```
Internal Router Routing Table
```

```
Codes: i - Intra-area route, I - Inter-area route
```

```
i 5.5.5.5 [2] via 192.168.2.2, GigabitEthernet1/0/1, ABR, Area 2,
SPF 3
```

Along with “show ipv6 ospf border-routers,” this command shows that there are multiple areas in an OSPF-routed network by the existence of a border router. A border router sits between two areas, which is only possible with multi-area OSPF.

```
MLS1#show ip ospf database router
```

```
OSPF Router with ID (7.7.7.7) (Process ID 7)
```

```
Router Link States (Area 2)
```

```
LS age: 1073
```

```
Options: (No TOS-capability, DC)
```

```
LS Type: Router Links
```

```
Link State ID: 5.5.5.5
```

```
Advertising Router: 5.5.5.5
```

```
LS Seq Number: 8000000D
```

```
Checksum: 0xD04A
```

```
Length: 36
```

```
Area Border Router
```

```
Number of Links: 1
```

```
Link connected to: a Transit Network
```

```
(Link ID) Designated Router address: 192.168.2.6
```

```
(Link Data) Router Interface address: 192.168.2.6
```

```
Number of MTID metrics: 0
```

```
TOS 0 Metrics: 1
```

Paired with “show ipv6 ospf database router,” this command reveals that OSPF has been configured with multiple areas given the label (Area 2) assigned to this particular MLS.

Problems

When starting this lab, after finishing the topology design, I tried to console into one of the routers, but the connection was unsuccessful. I tried other devices, but it still failed. To troubleshoot, I used an extra console cable to connect to various points in the system of wires. Eventually, the connection was successful, but the problem was still unclear. The wires were nested too deeply into the equipment rack

so I could not debug the exact issue. Instead, I continued to use a spare console cable to bypass the weak point.

After the configurations for the routers were finished, I tested a ping from Router0 to Router4, but it failed. I checked the IP addressing and noticed a duplicate IPv6 address use for two links on Router1 and Router2. After fixing this duplicate IPv6 address, I found that there were errors when pasting configurations into the routers. I noticed that the problems were caused by unnecessary lines in the config, so I removed excess lines, allowing a clean paste and making the config files neater.

However, when I tried to configure the MLSs, I found that they would not allow routing nor IP addresses on the interfaces. After researching, I learned that you must enable IP routing before beginning the configuration and that the command “no switchport” must be run on each interface.

Finally, when uploading my configurations to a new set of routers and switches, IPv6 stopped working. When uploading the configurations, the routers would throw the error message “OSPF Process <PID> failed to allocate unique router-id and cannot start.” To fix this, I tried making the OSPF and OSPFv3 process IDs different within the same router, but this did not solve the problem. So, I decided to reboot the routers and reupload the configurations, and the issue disappeared. The problem was never clear, but I suppose the configurations were either incorrectly pasted into the routers or the routers already had a configuration uploaded before I tried to upload mine.

Conclusion

OSPF is a powerful yet simple routing protocol that is easy to configure on any network. Its flexibility with multiple areas and the capability to route both IPv4 and IPv6 make it a strong choice for networks of any size. By dividing a network into areas, OSPF helps reduce the amount of routing information that each router needs to process, which makes the network more efficient and stable. This is especially useful in larger networks, where keeping routing tables small and limiting unnecessary updates can improve performance. Routing protocols are essential for making sure packets move across the network correctly, and OSPF provides a quick, reliable, and scalable option for doing so.